

## Continuous vs. Discrete - Practice

Algebra II | Unit 1 | Lesson 3 Practice | TEKS A2.2.A, A2.7.I

**Objective:** I can classify a domain or range as continuous or discrete, and write it using the notation that fits.

### Section 1 — Recall

- Continuous or discrete? (a) the volume of water poured from a jug (b) the number of keys on a keyboard.  
a. \_\_\_\_\_ b. \_\_\_\_\_
- A box volume is  $V(x) = x^3$ . Give the domain for the "built from 1-inch blocks" case (set notation) and the "any size box" case (interval notation).  
Blocks: { \_\_\_\_\_ } Any size: \_\_\_\_\_

### Section 2 — Apply

- A stadium sells hot dogs in whole units, 0 to 8 per person. Give the domain in set notation, and explain in one phrase why interval notation does not fit.  
Domain: { \_\_\_\_\_ }  
Why not interval: \_\_\_\_\_
- A candle 15 cm tall burns down to 0 cm at a constant rate. Give the domain of heights in interval notation, and classify it.  
Domain: \_\_\_\_\_ Classification: \_\_\_\_\_

### Section 3 — Challenge

- A school orders buses for a field trip; each bus holds 48 students, and 260 students are going. How many buses are needed, and why can't the answer be the raw division result?  
\_\_\_\_\_  
Buses needed: \_\_\_\_\_ Why: \_\_\_\_\_
- A 12-mile trail is tracked two ways: a GPS records a hiker's position continuously as she walks, while park rangers only log her position at each whole-mile marker. Explain why the first domain is continuous and the second is discrete, even though both describe the same hike.  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

### Section 4 — Explain It

- A classmate argues that because no real-world measurement is infinitely precise, all real-world distances should be treated as discrete, not continuous. Do you agree or disagree? Explain your reasoning.  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_